Android Introduction

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ICT Department, USTH

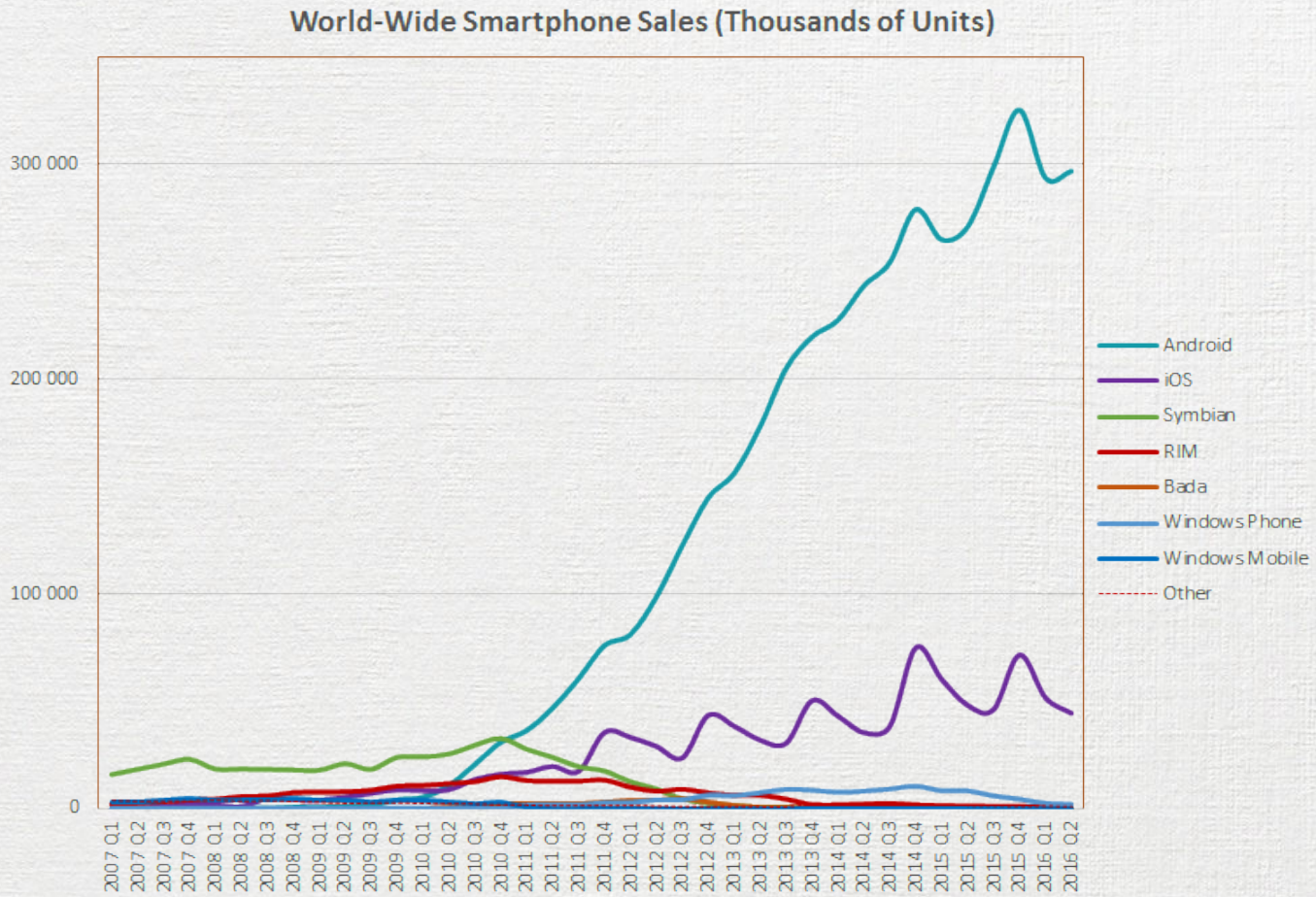
Introduction

Introduction

Period	Android	iOS	Windows Phone	Others
2015Q3	84.3%	13.4%	1.8%	0.5%
2015Q4	79.6%	18.6%	1.2%	0.5%
2016Q1	83.4%	15.4%	0.8%	0.4%
2016Q2	87.6%	11.7%	0.4%	0.3%

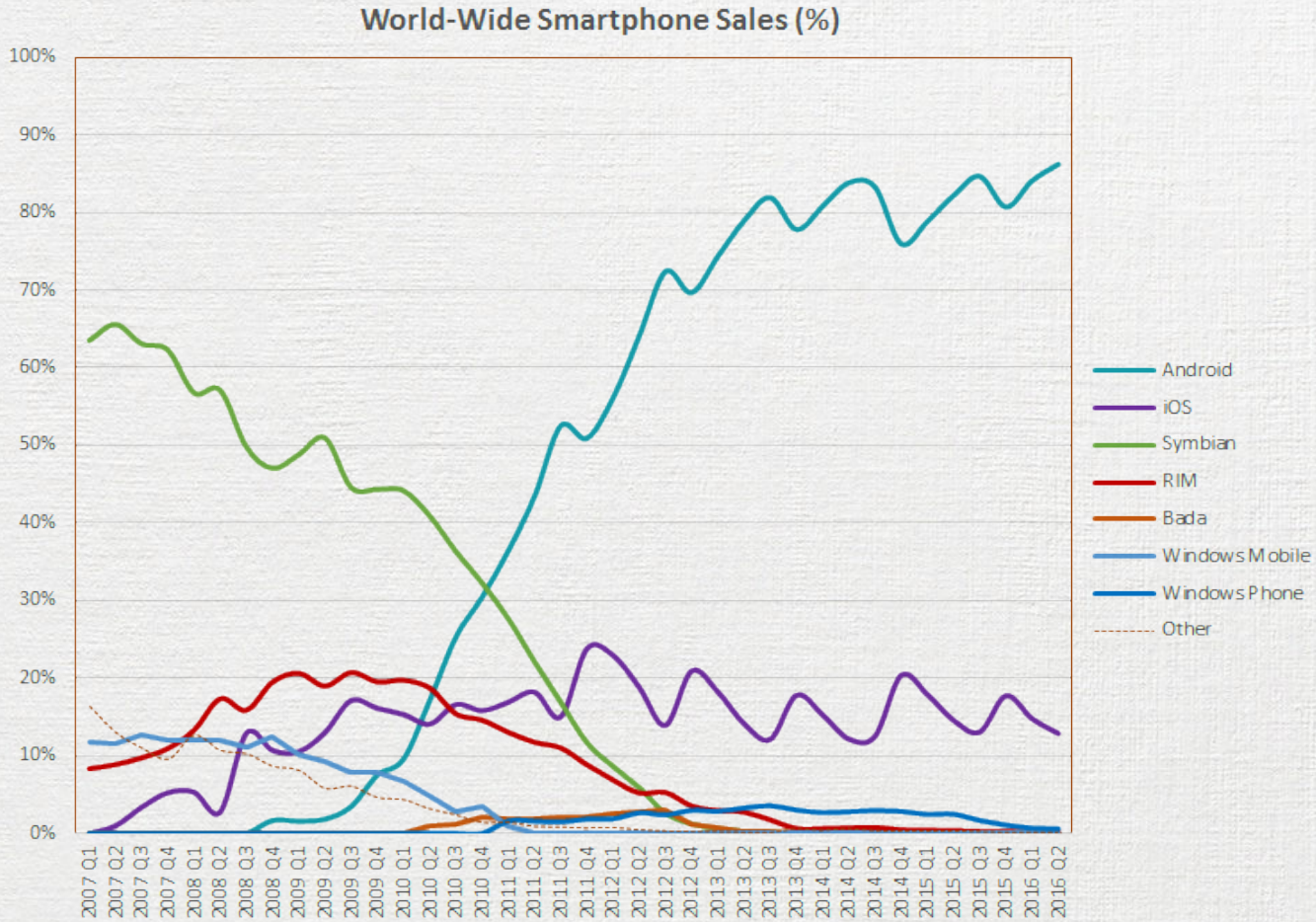
Source: IDG, Aug 2016

Introduction



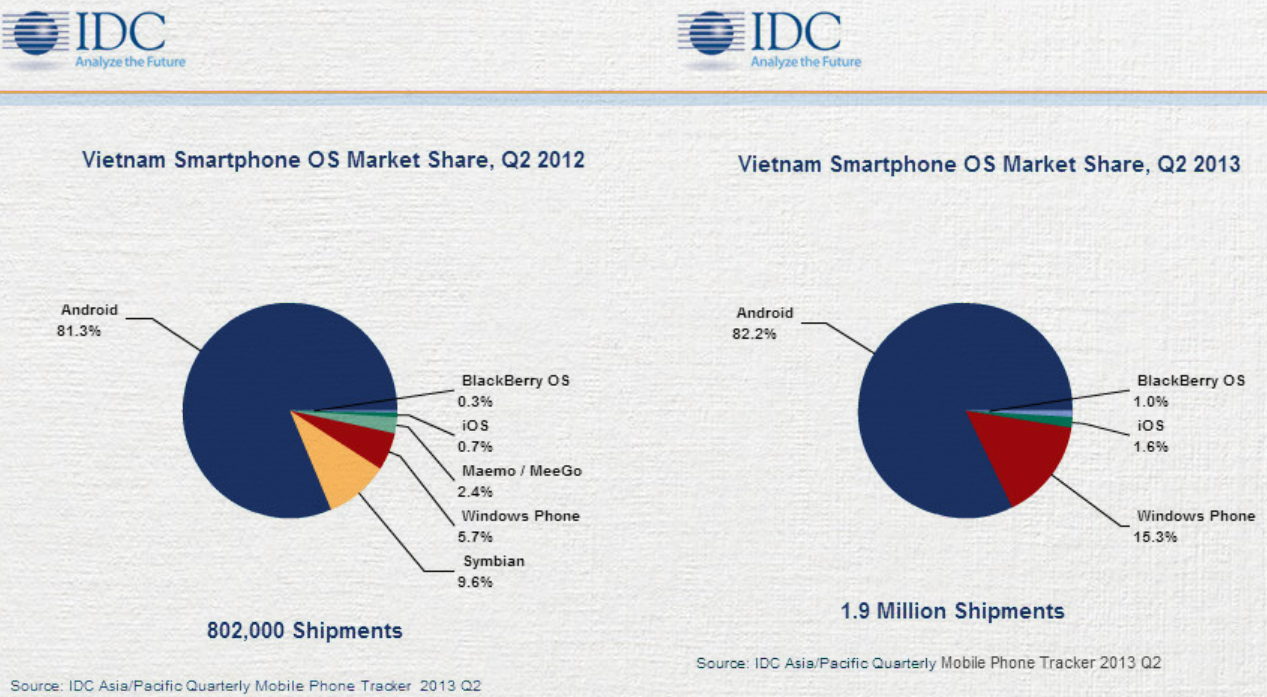
Source: Gartner & Wikipedia

Introduction



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Source: IDG, Mar 2016

Introduction

- Mobile Operating System
- Linux Distribution
- A set of user-space applications and frameworks
- Java
- **NOT** a device

Introduction

- Phones
- Tablets
- Watches
- TVs
- Set-top boxes

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- TVs
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- Cars
- Glasses
- Notebooks
- Game consoles
- Digital cameras

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  - ARM 32/64 bits
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- GPU
  - None
  - 1 - 550 GFLOPS

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  - Touchscreen
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- GSM/CDMA
- Camera
- WiFi
- Bluetooth
- Keyboard/Mouse

Why Android?

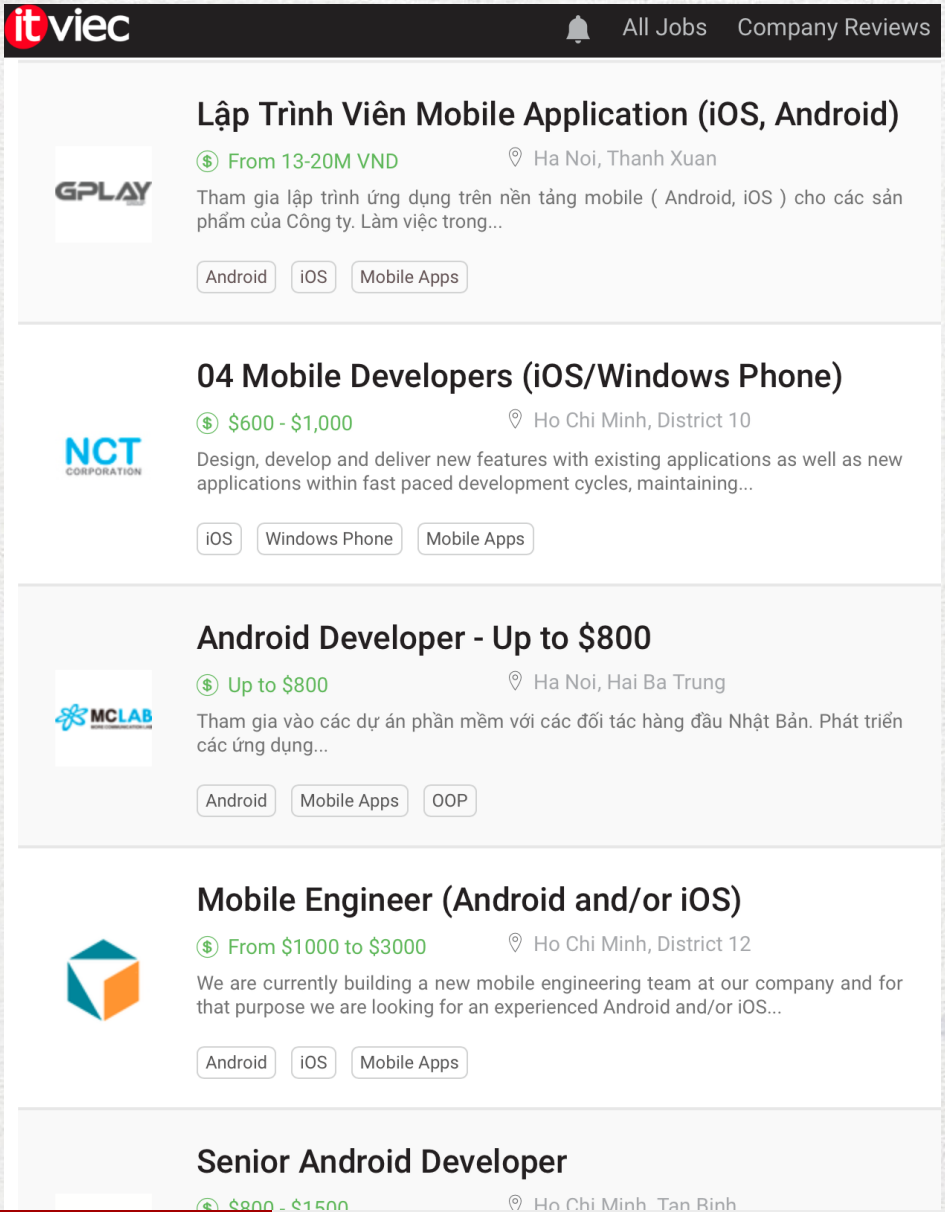
- Why not?
- Awesome!
- Popular
- Cheap
- Daily friend

Why Android?

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- Awesome!
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- Fast
- Easy to use
- Camera
- Network
- **Jobs!**

Why Android?


itviec
All Jobs
Company Reviews
Lập Trình Viên Mobile Application (iOS, Android)
From 13-20M VND
Ha Noi, Thanh Xuan
Tham gia lập trình ứng dụng trên nền tảng mobile (Android, iOS) cho các sản phẩm của Công ty. Làm việc trong...
Android
iOS
Mobile Apps
04 Mobile Developers (iOS/Windows Phone)
$600 - $1,000
Ho Chi Minh, District 10
Design, develop and deliver new features with existing applications as well as new applications within fast paced development cycles, maintaining...
iOS
Windows Phone
Mobile Apps
Android Developer - Up to $800
Up to $800
Ha Noi, Hai Ba Trung
Tham gia vào các dự án phần mềm với các đối tác hàng đầu Nhật Bản. Phát triển các ứng dụng...
Android
Mobile Apps
OOP
Mobile Engineer (Android and/or iOS)
From $1000 to $3000
Ho Chi Minh, District 12
We are currently building a new mobile engineering team at our company and for that purpose we are looking for an experienced Android and/or iOS...
Android
iOS
Mobile Apps
Senior Android Developer

Why Android?

- Freshmen will **NOT** have these salaries
- Good read (in Vietnamese):
  - https://toidicodedao.com/2016/09/29/coder-sieu-nhan/

Why Android?

- Reinforce the basics: OOP, decomposition, etc.
- Separation of UI design and functionality
- Asynchronous Programming
- Callback Methods

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Why not iOS in Bachelor?

- You can learn iOS in USTH ICT Master :-)

What is needed?

- OOP: package, inheritance, method overriding, aggregation, interface,. . .
- Java: casting, exceptions
- Networking: client-server model, HTTP, API
- Data representation: XML, JSON

What is needed?

- A laptop/desktop/workstation, at least:
  - Dual core CPU
  - 4GB memory
- [Optional, but recommended] An Android phone, with USB cable

What is needed?

- Any desktop OS: Windows, Linux, Mac OS X
- IDE
  - Android Studio [preferred]
  - Eclipse with Android Development Tools
  - Intellij IDEA
- Versioning Control System
  - git
- A good brain.

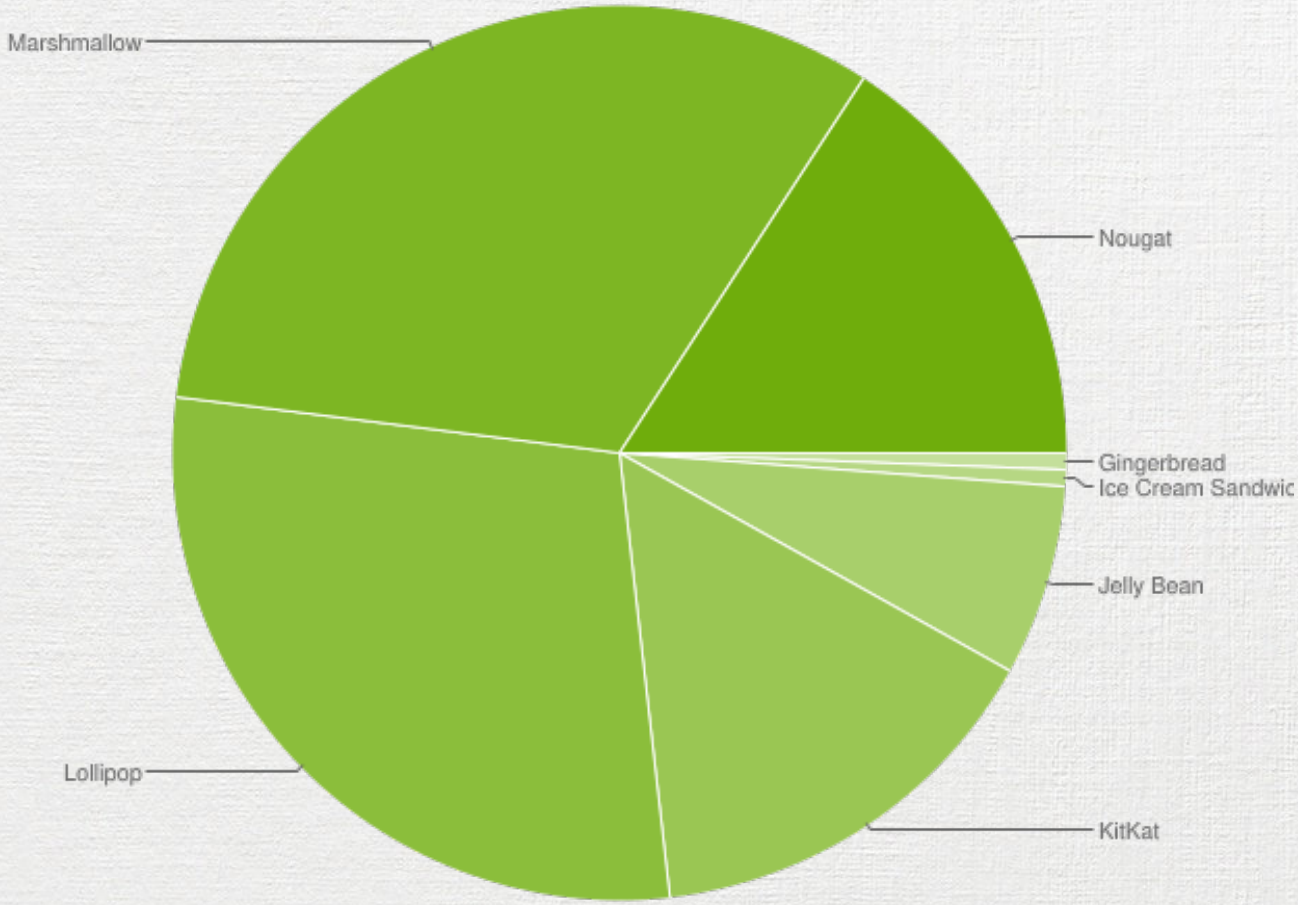
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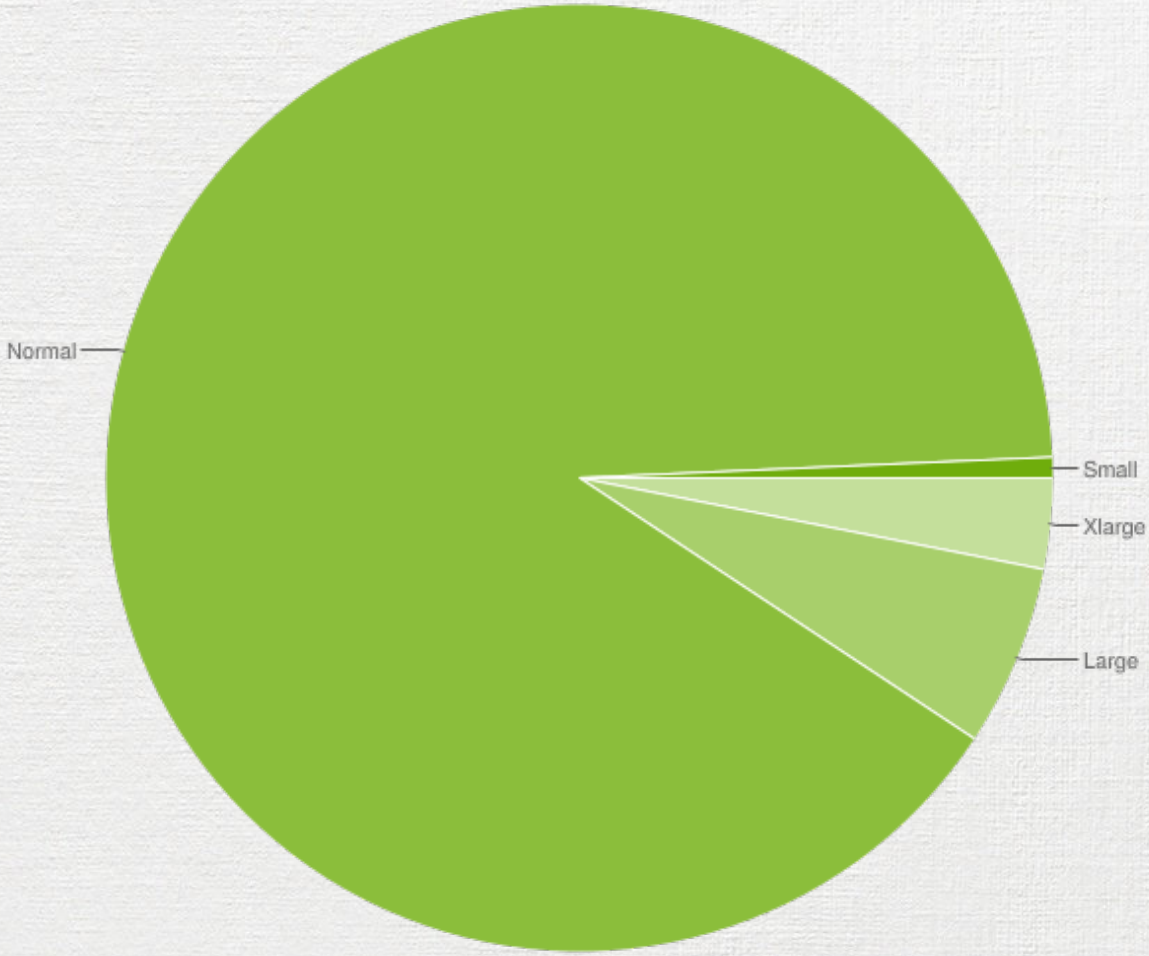
Timeline, Oct 2017

Facts



Platform Versions, Oct 2017

Facts



Device Sizes, Oct 2017